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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CCT)(CBCS) Degree Examinations

December 2024- January 2025

22CCT101 – CONSTRUCTION PLANNING AND CONTROL MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|--------------|------------|------------|------------|
| 1. a) How does formal organization structure differs from informal organization structure? | 8 | L*1 | 1 | 1 |
| b) Define the terms | | | | |
| i. Critical path and sub-critical path | | | | |
| ii. Earliest event time and Latest event time | 6 | L1 | 1 | 1 |
| iii. Total float and free float | | | | |
| c) How are the different levels of WBS classified? Explain with a neat flowchart and how does it facilitate communication among stakeholders? | 6 | L2 | 1 | 1,3 |
| 2. a) Explain the features of line and staff organization with the help of flow chart and list the merits and demerits. | 8 | L1 | 1 | 1 |
| b) Discuss the procedure of critical path determination. | 6 | L2 | 1 | 2 |
| c) Draw a bar chart for certain project whose activity breakdown is as follows
Activity B and C are done concurrently and both must follow activity A. Activity B should precede D. Activity E cannot start until both C and B are completed. Activity F and G can be started only after activities D and E are completed. What is the total time taken for project completion? | | | | |

Activity	A	B	C	D	E	F	G
Duration (weeks)	1	2	4	3	1	2	4

6 L3 1 2

Unit – II

3. a) Draw the network and number the nodes for the given precedence relations
- | | | | | | | | | | | | | |
|--------------------|---|---|---|---|---|------|---|---|---|---|------|------|
| Activity | A | B | C | D | E | F | G | H | I | J | K | L |
| Preceding Activity | - | A | B | A | D | C, E | D | D | H | H | F, H | G, J |
- b) Briefly explain the possible logic errors encountered while drawing network diagrams.
- c) Construct network diagram and compute
- Early start and finish times
 - Late start and finish times
 - Total float
 - Free float
 - Determine critical path (highlight it in network diagram)
 - Determine total project duration

Activity	1-2	2-3	2-4	3-5	3-6	4-5	4-7	5-8	6-8	7-8
Duration (days)	2	8	10	6	3	3	7	5	2	8

12 L3 2 2

4. a) Draw an activity on arrow diagram to represent the following project. Find the project's total duration and date of completion (assume the project start date is 1 /10/2024 and all days are working days). Calculate the occurrence of activity times and activity floats (Total float and free float). Also determine the critical path.

Activity	A	B	C	D	E	F	G	H	I	J	K
Duration (days)	4	5	3	2	5	3	4	5	5	6	7
Preceding Activity	-	-	-	A	B	B	C	D	E	H,I	F,G

12 L3 2 2
8 L1 2 1

- b) CPM and PERT being primary techniques used in network analysis how do they differ from one another?

Unit – III

5. a) Draw a PERT network and find expected project completion time. Variance and standard deviation.

What is the probability that the project finishes in less than 50 days?

Estimate the project completion time with atleast 95% confidence level. Take $z = 1.65$

Activity	(1-2)	(2-3)	(2-4)	(3-5)	(4-6)	(5-7)	(6-7)	(7-8)
Optimistic time	4	5	4	10	8	4	1	6
Most likely time	6	7	8	18	9	8	2	7
Pessimistic time	8	15	12	26	16	12	3	8

12 L3 2 2

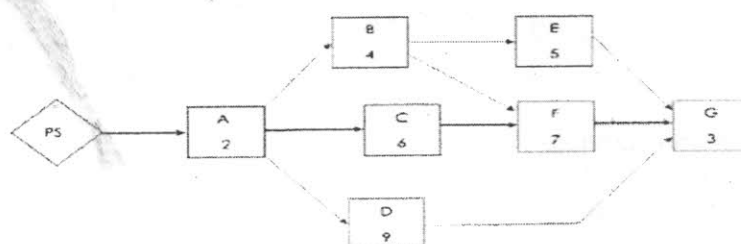
Explain

- b) I. Purpose of PERT methodology in project management
II. Purpose of line of balance scheduling?

8 L2 3 1

6. a) For the given network manually level the resources such that you may not deploy more than nine laborers per day at any time.

Activity	A	B	C	D	E	F	G
Laborers	3	2	4	3	4	2	4



12 L4 3 3

- b) What are the key steps taken in risk management and list some of the incidental risks associated with poor project management performance?

8 L2 3 2

Unit – IV

7. a) Define
I. Normal time
II. Crash time
III. Normal cost
IV. Crash cost

4 L1 4 1

- b) The following table gives the data for the duration and costs of each activity of a project network. The indirect cost of project is 3000/week. Determine the optimum duration of the project and the corresponding minimum cost. Draw the time scaled version of the network.

Activity	Normal duration (weeks)	Normal cost	Crash duration (weeks)	Crash cost
1-2	6	7000	3	14500
1-3	8	4000	5	8500
2-3	4	6000	1	9000
2-4	5	8000	3	15000
3-4	5	5000	3	11000

- c) What is the relation between project duration and project cost explain with the help of representation.

12	L4	4	3
4	L2	4	1

8. a) A company manufactures two types of products, P1 and P2. Each product uses a lathe and milling machine. The processing time for P1 on the lathe is five hours and on the milling machine it is four hours. The processing time for P2 on the lathe is 10 hours and on the milling machine it is four hours. The maximum number of hours per week available on the lathe and the milling machine are 60 hours and 40 hours, respectively. The profit per unit of selling P1 and P2 are Rs. 6.00 and Rs. 8.00, respectively. Formulate a linear programming model to determine the production volume of each product such that the total profit is maximized. A company manufactures two types of products, P1 and P2. Each product uses a lathe and milling machine. The processing time for P1 on the lathe is five hours and on the milling machine it is four hours. The processing time for P2 on the lathe is 10 hours and on the milling machine it is four hours. The maximum number of hours per week available on the lathe and the milling machine are 60 hours and 40 hours, respectively. The profit per unit of selling P1 and P2 are Rs. 6.00 and Rs. 8.00, respectively. Formulate a linear programming model to determine the production volume of each product such that the total profit is maximized.

4	L4	4	3
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- b) Solve the LPP by simplex method for optimum solution

Minimize $Z = 7x_1 + 5x_2$ subjected to constraints

$$x_1 + 2x_2 \leq 6$$

$$4x_1 + 3x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

8	L5	4	5
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- c) Solve the LPP by graphical method

Maximize $Z = 12x_1 + 16x_2$ subjected to constraints

$$10x_1 + 20x_2 \leq 120$$

$$8x_1 + 8x_2 \leq 80$$

$$x_1, x_2 \geq 0$$

8	L4	4	3
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Unit – V

9. a) Explain the objectives and functions of material management.
 b) What are the basic assumptions of EOQ model?
 c) Bob equipment company is in necessity of 48000 units of an equipment parts. Its holding cost is 15% with unit price of Rs.4 and the ordering cost is Rs. 9. Based on the given details suggest how many orders should be placed in a year and how often. Also mention the most economical number of units to order.

8	L2	5	1
4	L1	5	1

8	L3	5	3
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10. a) A construction site has limited storage space, what inventory strategies can be implemented to ensure smooth running of project without material shortages or excess?
- b) Explain different Inventory control techniques.
- c) Sarah Brown works for a manufacturer that makes parts for marine engines. The parts are made in batches and every time a new batch is started it costs Rs.1,640 for disruption and lost production and Rs.280 in wages for the fitters. One item has an annual demand of 1,250 units with a selling price of Rs.300, 60% of which is direct material and production costs. If the company looks for a return of 20% a year on capital, what is the optimal batch size for the item and the associated costs?

4	L2	5	2
8	L2	5	1
8	L3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
